

Bayesian Spatial Extreme Value Analysis of Precipitation in Málaga and Andalucía

Project Documentation and Model Specification

LGM_precipitation

[PhD Student]

Department of Statistics, University of Iceland

Jesús Vías

Department of Geology, University of Málaga, Spain

Draft begun 3 March 2026 — compiled March 5, 2026

Contents

I	Project Genesis and Design Decisions	3
1	How This Project Came About	3
2	The Problem: Extreme Precipitation in Málaga and Andalucía	3
2.1	Why This Matters	3
2.2	The DANA Phenomenon	4
3	Research Questions	4
4	Methodological Approach	4
4.1	Overview	4
4.2	Why Max-and-Smooth with PPP, Not INLA with GEV	5
4.3	Why Station Data, Not Gridded Products	5
5	Model Specification	6
5.1	Notation	6
5.2	Data Model: Poisson Point Process Likelihood	6
5.3	Latent Model and Parameterisation	6
5.4	Matérn(5/2) Gaussian Process	7
5.5	Return Levels	7
6	Data	7
6.1	Primary Source: AEMET Rain Gauge Network	7
6.2	Covariates	8
6.3	Supplementary Data Sources	8
II	Implementation	8
7	R Environment and Packages	9

8	Data Acquisition	9
9	Exploratory Analysis	10
10	Max-and-Smooth Model Fitting	11
10.1	Stage 1: Site-wise Poisson Point Process MLEs	11
10.2	Stage 2: MCMC Smoothing with Matérn GP (Stan)	11
10.3	Smoothing Effect	12
11	Results: Return Level Maps	12
A	Proposed extension: joint daily-GEV model	14
A.1	Motivation	14
A.2	Model structure	14
A.3	Implementation plan	15

Part I

Project Genesis and Design Decisions

1 How This Project Came About

This project is a collaboration between a PhD student in statistics (based in Iceland, specialising in Bayesian latent Gaussian models) and his nephew, a geology professor at the University of Málaga who works on hydrogeological hazards in southern Spain.

The motivation is direct: **rainfall, floods, and droughts** are problems of immediate concern to the people and infrastructure of Málaga and Andalucía. Recent DANA (Depresión Aislada en Niveles Altos) events have caused devastating floods in eastern Spain, making probabilistic extreme rainfall analysis a topic of urgent practical relevance.

Bayesian latent Gaussian models with spatial random effects are a powerful framework for modelling extremes of precipitation. The key reference is the final chapter of Hrafnkelsson (2023), which presents a BLGM for high-dimensional spatial extremes of daily rainfall — methodology developed within our own research group at the University of Iceland.

2 The Problem: Extreme Precipitation in Málaga and Andalucía

2.1 Why This Matters

Málaga province and the broader Andalucía region are characterised by a Mediterranean climate with pronounced seasonal contrasts: long dry summers and a wet season concentrated in autumn–winter. This climatology creates a dual hazard:

1. **Extreme rainfall and floods.** Short-duration, high-intensity precipitation events — often triggered by DANA (“gota fría”) systems — can deliver a significant fraction of the annual rainfall total in a matter of hours. The steep, short river basins draining the coastal sierras into the Mediterranean respond rapidly, producing flash floods with little warning time.
2. **Droughts.** Multi-year precipitation deficits are a recurring feature of the regional climate, with severe consequences for agriculture, municipal water supply, and aquifer recharge. The Guadalhorce and Guadalmedina basins serving Málaga are particularly vulnerable.

For both hazards, decision-makers need *probabilistic* answers: not just “how much rain can we expect?” but “how much rain, with what probability, and how confident are we?” Classical frequency analysis provides point estimates of return levels at individual stations but cannot:

- Borrow strength across nearby stations to improve estimates at data-sparse locations,
- Predict return levels at ungauged sites (spatial interpolation),
- Quantify the full posterior uncertainty in return level estimates,
- Incorporate covariates (elevation, distance to coast, orographic exposure) in a principled way.

A Bayesian spatial extreme value model addresses all four limitations.

2.2 The DANA Phenomenon

DANA events (historically called “gota fría”) occur when a pocket of cold air at upper levels (typically 500 hPa) becomes detached from the polar jet stream and drifts southward over the western Mediterranean. When this cold upper-level air mass encounters warm, moisture-laden air over the sea surface, the resulting instability drives intense convective precipitation.

TODO: Add specific recent DANA events affecting Málaga/Andalucía with dates and recorded precipitation totals. Jesús to provide local context.

3 Research Questions

If successful, this project will produce Bayesian return level maps with full posterior uncertainty for extreme rainfall across Málaga and Andalucía. Specifically, we aim to answer:

1. **Spatial return levels.** What are the 20-, 50-, and 100-year return levels for daily precipitation at any location in the study area, including ungauged sites?
2. **Uncertainty mapping.** How uncertain are these return level estimates? Where is uncertainty highest — and where might additional monitoring stations reduce it most?
3. **Drivers of spatial variation.** How do the GEV parameters (location μ , scale σ , shape ξ) depend on geographic covariates such as elevation, distance to coast, and orographic exposure?
4. **Infrastructure adequacy.** How do estimated return levels compare against the rainfall thresholds used for drainage design, dam sizing, and civil protection planning in Málaga?
5. **Benchmark against classical methods.** How much does the Bayesian spatial model improve upon traditional at-site GEV fits that ignore spatial structure?

4 Methodological Approach

4.1 Overview

We adopt a **Bayesian latent Gaussian model** for spatial extremes, following the framework of Hazra et al. (2023). Instead of annual block maxima, we model **daily threshold exceedances** via the Poisson point process (PPP) likelihood, which provides a unified framework for both the exceedance rate and the tail behaviour.

Inference proceeds via the **Max-and-Smooth** algorithm (Hrafnkelsson et al., 2021):

Stage 1 (Max) At each station, fit the Poisson point process model independently to obtain MLEs of the transformed GEV parameters (ψ_i, τ_i, ϕ_i) and their covariance matrices (via parametric bootstrap).

Stage 2 (Smooth) Treat the site-wise MLEs as data in a Gaussian surrogate likelihood model with latent Gaussian spatial effects (Matérn GP). Fit via MCMC (Stan/NUTS) to obtain spatially smoothed posteriors for all parameters at every station.

The three-level hierarchical structure is:

Data layer Daily precipitation threshold exceedances at each station, modelled by the PPP likelihood with station-specific GEV parameters (μ_i, σ_i, ξ_i) .

Latent layer	Transformed parameters $\psi_i = \log(\mu_i)$, $\tau_i = \log(\sigma_i/\mu_i)$, $\phi_i = h(\xi_i)$ are linked to intercepts plus Matérn(5/2) Gaussian process spatial random effects and nuggets (for all three parameters).
Prior layer	Penalised complexity (PC) priors (Simpson et al., 2017; Fuglstad et al., 2019) on the GP hyperparameters (range, marginal standard deviation) and nugget standard deviations.

4.2 Why Max-and-Smooth with PPP, Not INLA with GEV

Two inference strategies are available within the BLGM framework:

1. **R-INLA + GEV block maxima** (Rue et al., 2009): Fits a hierarchical GEV model on annual maxima in a single coherent inference step. Uses only one observation per station per year.
2. **Max-and-Smooth + PPP** (Hrafnkelsson et al., 2021; Hazra et al., 2023): Two-stage inference on daily threshold exceedances via the Poisson point process likelihood. Stage 1 computes site-wise MLEs; Stage 2 smooths them spatially via MCMC.

We choose **Max-and-Smooth with the PPP likelihood** because:

- The PPP likelihood uses *all* daily exceedances above a threshold, not just one observation per year. This extracts far more information from the tail and yields more precise estimates, especially for the shape parameter ξ .
- The PPP provides a unified model for both the exceedance rate and the tail behaviour (unlike a GP + Bernoulli split, which would require two separate models).
- R-INLA does not directly support the PPP likelihood; M&S provides a natural MCMC-based inference framework for it.
- The methodology follows exactly Chapter 7 of Hrafnkelsson (2023), providing a direct application of research from our group to a new domain.

Design Note: An INLA-based analysis using annual block maxima (GEV likelihood) could serve as a complementary comparison. The M&S + PPP approach is preferred as the primary method due to its more efficient use of the data.

4.3 Why Station Data, Not Gridded Products

Several gridded precipitation products exist for Spain (AEMET ROCIO, SPREAD, Spain02, Iberia01, ERA5-Land). However, all gridded products involve interpolation or model-based estimation, which tends to **smooth the tails** of the precipitation distribution — exactly the part we care about for extreme value analysis.

- **AEMET ROCIO** (5 km): Optimal interpolation of station data. Grid cell values are statistical estimates, not direct readings.
- **SPREAD** (5 km): Geostatistical interpolation with uncertainty estimates. Better than ROCIO for our purposes, but still smoothed.
- **ERA5 / ERA5-Land**: Reanalysis (numerical weather model output). Documented to underestimate heavy (≥ 20 mm/day) and violent (≥ 40 mm/day) precipitation over Spain.

We therefore use **raw AEMET rain gauge measurements** as our response variable. The spatial GP in the Bayesian model handles interpolation to ungauged locations in a principled way, without pre-smoothing the extremes.

Gridded products may still serve as covariates or for spatial context (e.g., ERA5 large-scale atmospheric fields as predictors of local extremes).

5 Model Specification

5.1 Notation

Let y_{ij} denote the j -th daily precipitation exceeding a threshold u_i at station i ($i = 1, \dots, n$), and let n_i^{ex} denote the number of exceedances at station i over n_T total observation days. Let $\mathbf{s}_i \in \mathbb{R}^2$ denote the spatial coordinates of station i .

5.2 Data Model: Poisson Point Process Likelihood

Daily precipitation exceedances follow a Poisson point process (PPP) model. The PPP log-likelihood for station i with threshold u_i , GEV parameters (μ_i, σ_i, ξ_i) , and $n_{py} = 365.25$ days per year is:

$$\ell_i(\mu_i, \sigma_i, \xi_i) = -n_i^{\text{ex}} \log \sigma_i - \left(\frac{1}{\xi_i} + 1 \right) \sum_{j=1}^{n_i^{\text{ex}}} \log \left[1 + \xi_i \frac{y_{ij} - \mu_i}{\sigma_i} \right] - \frac{n_T}{n_{py}} \left[1 + \xi_i \frac{u_i - \mu_i}{\sigma_i} \right]^{-1/\xi_i} \quad (1)$$

where:

- $\mu_i \in \mathbb{R}$ is the GEV location parameter,
- $\sigma_i > 0$ is the GEV scale parameter,
- $\xi_i \in \mathbb{R}$ is the GEV shape parameter ($\xi > 0$: heavy-tailed Fréchet; $\xi = 0$: Gumbel; $\xi < 0$: bounded Weibull),
- u_i is the station-specific threshold (we use the 75th percentile of positive daily precipitation at each station).

The PPP unifies the exceedance rate and tail behaviour in a single likelihood: the third term in (1) governs the expected number of exceedances per year, while the second term models the tail of the exceedance magnitudes. The GEV parameters (μ_i, σ_i, ξ_i) map directly to those of the annual maxima distribution.

Observations at different stations are modelled as **conditionally independent** given the GEV parameters; spatial dependence enters through the latent level.

Design Note: This conditional independence assumption is standard for marginal return level estimation. Modelling joint extremes would require max-stable or r-Pareto processes — a possible extension.

5.3 Latent Model and Parameterisation

Following Hazra et al. (2023), the GEV parameters are reparameterised as:

$$\psi_i = \log(\mu_i) \quad (2)$$

$$\tau_i = \log(\sigma_i/\mu_i) \quad (3)$$

$$\phi_i = h(\xi_i) = a_\phi + b_\phi \log(-\log(1 - (\xi_i + 0.5)^{c_\phi})) \quad (4)$$

where $c_\phi = 0.8$, and a_ϕ, b_ϕ are constants chosen so that h maps $\xi \in (-0.5, 0.5)$ to $\phi \in \mathbb{R}$, with inverse $g(\phi) = [1 - \exp(-\exp((\phi - a_\phi)/b_\phi))]^{1/c_\phi} - 0.5$. The bounded transformation prevents pathologically heavy or bounded tails.

The latent model for the transformed parameters is:

$$\psi_i = \beta_\psi + w_\psi(\mathbf{s}_i) + \varepsilon_i^\psi, \quad \varepsilon_i^\psi \sim N(0, \nu_\psi^2) \quad (5)$$

$$\tau_i = \beta_\tau + w_\tau(\mathbf{s}_i) + \varepsilon_i^\tau, \quad \varepsilon_i^\tau \sim N(0, \nu_\tau^2) \quad (6)$$

$$\phi_i = \beta_\phi + w_\phi(\mathbf{s}_i) + \varepsilon_i^\phi, \quad \varepsilon_i^\phi \sim N(0, \nu_\phi^2) \quad (7)$$

where:

- $\beta_\psi, \beta_\tau, \beta_\phi$ are intercepts,
- $w_\psi(\cdot), w_\tau(\cdot), w_\phi(\cdot)$ are zero-mean Gaussian process spatial random effects with Matérn(5/2) covariance (Section 5.4),
- $\nu_\psi^2, \nu_\tau^2, \nu_\phi^2$ are nugget variances.

Design Note: The book chapter (Hazra et al., 2023) models ϕ with intercept + nugget only (no spatial GP), arguing that ξ shows little spatial structure. We include a GP for ϕ as well, letting the data decide. The posterior confirms weak spatial dependence for ϕ : $\hat{\sigma}_\phi = 0.12$ (small marginal SD) and $\hat{\rho}_\phi = 3.6^\circ$ (very long range, effectively near-constant), consistent with the book’s finding but achieved without imposing the constraint *a priori*.

5.4 Matérn(5/2) Gaussian Process

Each spatial random effect $w_p(\cdot)$ ($p \in \{\psi, \tau, \phi\}$) is a zero-mean Gaussian process with a Matérn covariance function of smoothness $\nu = 5/2$:

$$C_p(\mathbf{s}, \mathbf{s}') = \sigma_p^2 \left(1 + \frac{\sqrt{5}d}{\rho_p} + \frac{5d^2}{3\rho_p^2} \right) \exp\left(-\frac{\sqrt{5}d}{\rho_p} \right) \quad (8)$$

where $d = \|\mathbf{s} - \mathbf{s}'\|$ is the Euclidean distance (in degrees), σ_p is the marginal standard deviation, and ρ_p is the practical range parameter.

The joint prior on the spatial effects at the $n = 127$ station locations is:

$$\mathbf{w}_p = (w_p(\mathbf{s}_1), \dots, w_p(\mathbf{s}_n))^\top \sim N(\mathbf{0}, \mathbf{K}_p) \quad (9)$$

where $[\mathbf{K}_p]_{ij} = C_p(\mathbf{s}_i, \mathbf{s}_j)$. We use a non-centred parameterisation $\mathbf{w}_p = \mathbf{L}_p \boldsymbol{\eta}_p^{\text{raw}}$, $\boldsymbol{\eta}_p^{\text{raw}} \sim N(\mathbf{0}, \mathbf{I})$, where $\mathbf{L}_p$ is the Cholesky factor of $\mathbf{K}_p$, to improve MCMC sampling efficiency.

Design Note: The original book chapter (Hazra et al., 2023) uses the SPDE approach of Lindgren et al. (2011) with Matérn($\nu = 1$) fields discretised on a triangular mesh, yielding a GMRF representation. Our direct GP approach with Matérn(5/2) operates on the 127×127 dense covariance matrix, which is feasible at this problem size and avoids mesh-construction choices. The higher smoothness ($\nu = 5/2$ vs $\nu = 1$) produces smoother spatial fields, which is appropriate for slowly-varying climatological parameters.

5.5 Return Levels

The p -year return level at station i (or any predicted location) is the quantile:

$$z_p(\mathbf{s}_i) = \mu_i + \frac{\sigma_i}{\xi_i} \left[\left(-\log\left(1 - \frac{1}{p}\right) \right)^{-\xi_i} - 1 \right] \quad (10)$$

Because (μ_i, σ_i, ξ_i) are functions of latent Gaussian variables with full posterior distributions, we obtain posterior distributions for $z_p(\mathbf{s}_i)$ at every location — including credible intervals and exceedance probabilities.

6 Data

6.1 Primary Source: AEMET Rain Gauge Network

Daily precipitation data are obtained from the Agencia Estatal de Meteorología (AEMET) via the OpenData API, accessed through the `climaemet` R package (climaemet, 2026). Key details:

- **Access:** Free API key from <https://opendata.aemet.es/centrodedescargas/obtencionAPIKey>
- **Variables:** Daily 24-hour accumulated precipitation (mm), station coordinates, elevation, province
- **Coverage:** ~947 stations across Spain; to be filtered for Andalucía (provinces: Málaga, Sevilla, Córdoba, Granada, Jaén, Almería, Cádiz, Huelva)
- **Record length:** Oldest stations from ~1960; no minimum record length is imposed — the Bayesian spatial model borrows strength across stations, so even short-record stations contribute useful spatial information
- **Quality control:** Basic QC provided by AEMET; we apply additional checks (negative values, implausible extremes, excessive missing data)

6.2 Covariates

Candidate spatial covariates for the latent model:

1. **Elevation** (m a.s.l.) — from station metadata or a DEM (e.g., SRTM 90 m, EU-DEM 25 m)
2. **Distance to coast** (km) — computed from station coordinates and a Mediterranean/Atlantic coastline shapefile
3. **Orographic index** — a measure of topographic exposure to moisture-bearing winds; computed from the DEM as the mean gradient in the dominant wind direction
4. **Longitude / latitude** — to capture large-scale east–west and north–south gradients in precipitation regime

TODO: Jesús to advise on additional covariates relevant to Málaga’s microclimate: aspect relative to the Cordillera Bética, distance to the Strait of Gibraltar, etc.

6.3 Supplementary Data Sources

AEMET ROCIO (5 km grid, 1951–present)

Official AEMET gridded product, optimal interpolation from 3,236 stations. Useful for spatial context and cross-validation, not as the primary response.

SPREAD (5 km grid, 1950–2012)

Geostatistical interpolation with per-cell uncertainty estimates. Useful for comparison and uncertainty propagation studies.

ERA5-Land (9 km, 1950–present)

ECMWF reanalysis. Underestimates extremes over Spain, but valuable as a source of large-scale atmospheric covariates (e.g., moisture flux, synoptic typing for DANA identification).

SAIH Hidrosur

Real-time automatic rain gauges for the internal basins of Andalucía. Useful for validating recent extreme events but shorter records.

Part II

Implementation

7 R Environment and Packages

```
# ---- Data acquisition ----
library(climaemet)      # AEMET OpenData API wrapper
library(sf)             # spatial features
library(terra)          # raster handling

# ---- Modelling (Max-and-Smooth) ----
library(cmdstanr)       # Stan interface (Stage 2 MCMC)
library(INLA)           # mesh construction (Stage 1 only)
library(Matrix)         # sparse matrix algebra
library(spam)           # sparse Cholesky decomposition
library(fields)         # distance matrices
library(coda)           # MCMC diagnostics
library(evd)            # classical GEV fitting (baseline)

# ---- Visualisation ----
library(ggplot2)
library(viridis)
library(patchwork)

# ---- General ----
library(tidyverse)
library(lubridate)
```

8 Data Acquisition

```
# ---- Step 1: Get AEMET station inventory ----
# Requires API key: aemet_api_key("YOUR_KEY", install = TRUE)

stations <- aemet_stations()

# Filter for Andalusia provinces
andalucia_provs <- c("MALAGA", "SEVILLA", "CORDOBA", "GRANADA",
                    "JAEN", "ALMERIA", "CADIZ", "HUELVA")

stations_and <- stations %>%
  filter(provincia %in% andalusia_provs) %>%
  st_as_sf(coords = c("longitud", "latitud"), crs = 4326)

cat("Stations in Andalusia:", nrow(stations_and), "\n")
```

```

# ---- Step 2: Download daily precipitation ----
# NOTE: AEMET API is rate-limited (50 requests/min).
# Download in batches by station and date range.

# Example: download for all Malaga stations, 1980-2024
malaga_daily <- aemet_daily_clim(
  station = stations_and$indicativo[stations_and$provincia == "MALAGA"],
  start   = "1980-01-01",
  end     = "2024-12-31",
  return_sf = TRUE
)

# ---- Step 3: Compute annual block maxima ----
annual_max <- daily_all %>%
  mutate(year = year(fecha)) %>%
  group_by(indicativo, year) %>%
  summarise(
    max_prec = max(prec, na.rm = TRUE),
    n_obs    = sum(!is.na(prec)),
    .groups  = "drop"
  ) %>%
  # QC: require 90% completeness and finite max
  filter(n_obs >= 330, is.finite(max_prec), max_prec >= 0)

# No minimum-year threshold: the Bayesian spatial model borrows
# strength across stations, so even short-record stations contribute.

cat("Stations:", n_distinct(annual_max$indicativo), "\n")
cat("Total station-year observations:", nrow(annual_max), "\n")

```

The acquisition scripts (R/01_acquire_aemet_data.R) have been run. Key results:

- 1,067,851 daily precipitation records downloaded (1960–2025)
- 127 stations across all 8 Andalusian provinces after QC (Almería 16, Cádiz 15, Córdoba 15, Granada 13, Huelva 15, Jaén 13, Málaga 24, Sevilla 16)
- No minimum record length imposed: the Bayesian spatial model borrows strength

9 Exploratory Analysis

EDA is implemented in R/02_exploratory_analysis.R and produces 8 diagnostic figures. Key findings:

- All at-site GEV shape parameters $\hat{\xi} > 0$ (Fréchet-type, heavy tails), ranging from 0.04 (Sevilla, Granada) to 0.27 (Córdoba)
- Strong spatial gradients: Málaga Airport has the highest $\hat{\mu} = 56$ mm; Grazalema (Cádiz) is the overall wettest site
- Empirical variogram of annual maxima shows clear spatial dependence up to ~ 200 km
- Positive elevation–maxima relationship, consistent with orographic enhancement

10 Max-and-Smooth Model Fitting

The M&S analysis is implemented in two stages: Stage 1 in `R/03_max_and_smooth.R` using functions from the companion code to Hazra et al. (2023) (`vendor/max_and_smooth/`), and Stage 2 in `R/03d_smooth_matern_pc.R` using a custom Stan program (`Stan/smooth_matern_pc.stan`).

10.1 Stage 1: Site-wise Poisson Point Process MLEs

For each of the 127 stations, we:

1. Set the threshold u_i as the 75th percentile of positive daily precipitation (range: 3.0–23.9 mm, median 8.4 mm).
2. Maximise the PPP log-likelihood (1) to obtain MLEs $(\hat{\psi}_i, \hat{\tau}_i, \hat{\phi}_i)$.
3. Compute the covariance matrix of the estimators via 1,000 parametric bootstrap samples.

All 127 stations converge. The MLEs on the original scale are:

- $\hat{\mu}$: 20.0–131.0 mm (median 42.6 mm)
- $\hat{\sigma}$: 6.6–36.9 mm (median 13.5 mm)
- $\hat{\xi}$: –0.026 to 0.312 (median 0.104)

10.2 Stage 2: MCMC Smoothing with Matérn GP (Stan)

Stage 2 fits the Gaussian surrogate likelihood to the Stage 1 MLEs, incorporating Matérn(5/2) GP spatial random effects for all three transformed parameters (ψ, τ, ϕ) . The model is implemented in Stan (Carpenter et al., 2017) using the No-U-Turn Sampler (NUTS):

- 4 chains $\times$ 2,000 iterations (1,000 warmup + 1,000 sampling) = 4,000 posterior draws
- `adapt_delta` = 0.9, `max_treedepth` = 12
- Non-centred parameterisation for the GP (see Section 5.4)
- Runtime: $\sim$ 130 minutes on 4 CPU cores
- Divergences: 112/4,000 (3%), spread evenly across chains; all $\hat{R} \leq 1.01$

PC priors. We use penalised complexity priors (Simpson et al., 2017; Fuglstad et al., 2019) on all hyperparameters, calibrated via probability statements:

- GP marginal SD: $\sigma_p \sim \text{Exp}(\lambda_\sigma)$ with $\lambda_\sigma = 3.0$, i.e. $P(\sigma_p > 1) = 0.05$
- GP range: $\pi(\rho_p) = \lambda_\rho \rho_p^{-2} \exp(-\lambda_\rho/\rho_p)$ with $\lambda_\rho = 0.30$, i.e. $P(\rho_p < 0.1^\circ) = 0.05$
- Nugget SD: $\nu_p \sim \text{Exp}(\lambda_\nu)$ with $\lambda_\nu = 6.0$, i.e. $P(\nu_p > 0.5) = 0.05$
- Intercepts: $\beta_p \sim N(0, 100^2)$ (weakly informative)

Posterior hyperparameter summaries:

Parameter	Posterior mean	95% CI	Interpretation
β_ψ	3.67	(3.41, 3.87)	Intercept for $\log(\mu)$
β_τ	-1.04	(-1.33, -0.66)	Intercept for $\log(\sigma/\mu)$
β_ϕ	0.18	(0.01, 0.49)	Intercept for shape
σ_ψ	0.26	(0.18, 0.41)	GP marginal SD for ψ
σ_τ	0.23	(0.12, 0.49)	GP marginal SD for τ
σ_ϕ	0.12	(0.03, 0.40)	GP marginal SD for ϕ
ρ_ψ	0.70	(0.45, 1.07)	GP range for ψ (degrees)
ρ_τ	1.65	(0.99, 2.75)	GP range for τ (degrees)
ρ_ϕ	3.61	(0.98, 10.88)	GP range for ϕ (degrees)
ν_ψ	0.150	(0.125, 0.179)	Nugget SD for ψ
ν_τ	0.011	(0.002, 0.019)	Nugget SD for τ
ν_ϕ	0.003	(<0.001, 0.007)	Nugget SD for ϕ

Design Note: The GP range for ϕ ($\rho_\phi = 3.6^\circ$, 95% CI: 0.98–10.9°) is poorly constrained, with a posterior that wanders to values exceeding the study domain diameter ($\sim 6^\circ$). Combined with the small marginal SD ($\sigma_\phi = 0.12$), this confirms that the shape parameter has minimal spatial structure — the GP for ϕ is effectively fitting a near-constant surface. This is consistent with the book chapter’s decision to omit the spatial GP for ϕ .

Prior sensitivity. We also fitted the model with weakly informative priors (Inverse-Gamma for range, Exponential for SD) instead of PC priors. The posteriors for ψ and τ are virtually identical under both prior specifications; only the weakly-identified ϕ hyperparameters shift. Station-level 100-year return levels differ by < 1 mm, confirming that the results are robust to the choice of prior. The PC prior model is retained as the primary specification due to its principled construction and connection to the reference methodology.

10.3 Smoothing Effect

Spatial smoothing substantially reduces the variability of the parameter estimates compared to the site-wise MLEs:

- μ : MLE range 20.0–131.0 $\rightarrow$ smoothed 20.8–128.7 mm
- σ : MLE range 6.6–36.9 $\rightarrow$ smoothed 7.3–38.6 mm
- ξ : MLE range -0.026 to 0.312 $\rightarrow$ smoothed 0.065 to 0.215

The strongest smoothing effect is on the shape parameter, where 127 noisy site-wise estimates are pulled toward a common value of $\xi \approx 0.10$ (median 0.097), consistent with the relatively homogeneous heavy-tailed (Fréchet-type) precipitation regime across Andalucía.

11 Results: Return Level Maps

Return levels are computed from the posterior chains using equation (10).

Return period	Range of posterior mean (mm)	Range of posterior SD (mm)
20-year	48.4 – 260.2	2.1 – 12.1
50-year	59.2 – 309.3	2.8 – 16.8
100-year	68.3 – 348.9	3.5 – 21.5

The Mediterranean coast shows the highest extremes overall, while inland Granada and semi-arid Almería have the lowest. By province, median 100-year return levels are: Málaga (158 mm), Cádiz (136 mm), Huelva (129 mm), Sevilla (116 mm), Almería (104 mm), Córdoba (103 mm), Jaén (98 mm), Granada (97 mm).

Figure 1 shows the posterior predictive interpolation maps for the GEV parameters and 100-year return level, produced via kriging on a 0.025° grid covering Andalucía.

Posterior predictive: extreme precipitation across Andalucía (PC priors)

Matérn(5/2) GP + PC priors | 127 stations, 14278 grid points, 4000 draws | 0.025° tiles

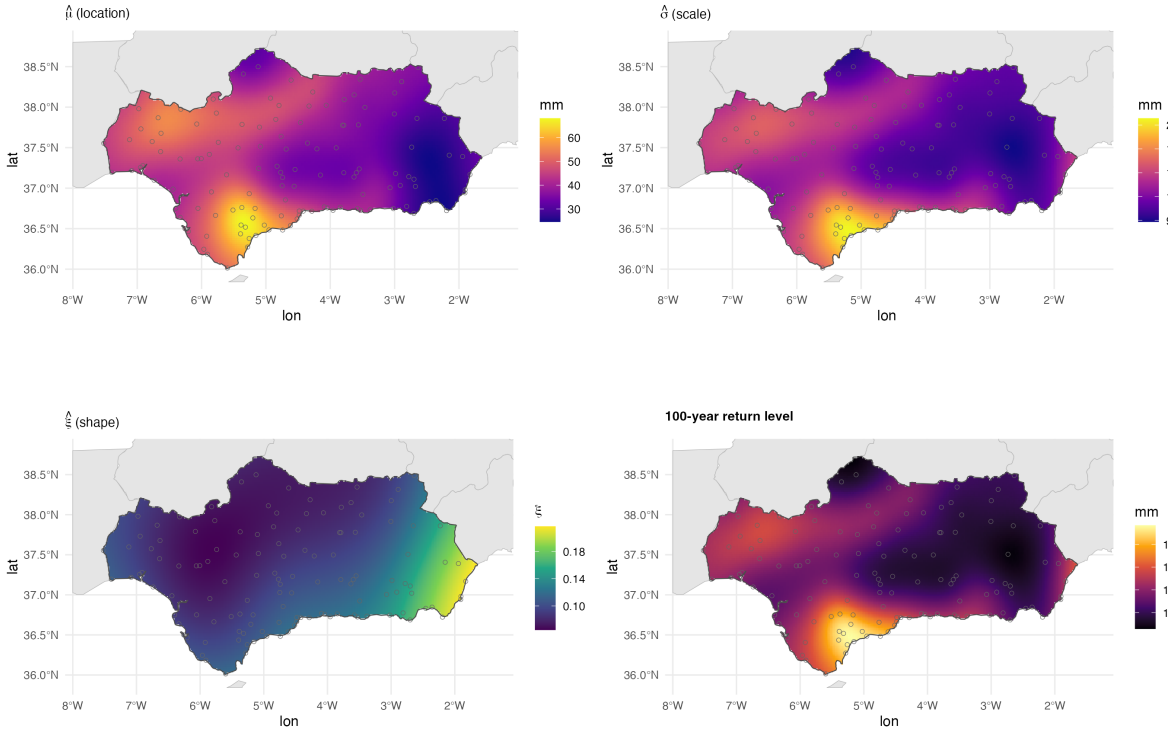


Figure 1: Posterior predictive interpolation maps (Matérn GP + PC priors). Top row: GEV location ($\hat{\mu}$, mm), scale ($\hat{\sigma}$, mm), and shape ($\hat{\xi}$). Bottom row: 100-year return level (posterior mean, mm), posterior SD (mm), and coefficient of variation. Hollow circles indicate station locations.

TODO: Extend the discussion:

- Compare M&S return levels against at-site GEV fits at long-record stations to quantify the benefit of spatial borrowing
- Cross-validation: leave-one-station-out predictive check
- Add covariates (elevation, distance to coast) to the latent model
- Discuss implications for infrastructure design thresholds in Málaga
- Relax completeness filter (330 $\rightarrow$ 280 days) with season-aware criterion; 19 rejected station-years have annual max > 100 mm
- Non-stationary GEV: time-varying $\mu(t)$ or $\sigma(t)$ to account for climate trends (current model assumes stationarity 1960–2025)

A Proposed extension: joint daily–GEV model

The current Max-and-Smooth analysis treats annual block maxima as the fundamental data unit. This appendix outlines a proposed extension that models *daily* rainfall directly, enabling principled imputation of missing observations and a tighter link between the daily rainfall process and the extreme value model.

A.1 Motivation

Our 90% completeness filter (requiring ≥ 330 daily observations per station-year) discards 528 of 3029 station-years (17%), including 19 with annual maxima exceeding 100 mm. Analysis of the missing days shows that gaps are distributed across the calendar year (not concentrated in summer), so some informative extreme events may be lost. The core question is: if a station was offline during October (peak DANA season), what might we be missing?

A secondary motivation is that the GEV shape parameter ξ — which governs tail heaviness and is our most uncertain parameter — is currently estimated from ~ 2500 annual maxima. A daily model would inform ξ from $\sim 1\,000\,000$ daily observations, dramatically sharpening the tail estimate.

A.2 Model structure

The proposed model has a modular structure with a one-directional dependency: the daily model is estimated first (or jointly), and the GEV emerges as a derived consequence.

Layer 1: Daily rainfall process. For station s on day t , daily precipitation follows a hurdle model:

$$P(Y_{s,t} > 0) = \text{logit}^{-1}(\alpha_0(s) + f_p(d(t), s)), \quad (11)$$

$$Y_{s,t} \mid Y_{s,t} > 0 \sim \text{GPD}(\sigma(s, d(t)), \xi(s)) \quad \text{for } Y_{s,t} > u(s), \quad (12)$$

where $d(t)$ is the day-of-year (1–365), $f_p(\cdot)$ is a smooth seasonal function (e.g. Fourier harmonics or cyclic spline), $u(s)$ is a high threshold (e.g. the 95th percentile of wet-day amounts), and the Generalized Pareto Distribution (GPD) models exceedances above u .

Below the threshold, positive rainfall amounts can be modelled by a simpler distribution (e.g. Gamma), since only the tail matters for extremes.

Layer 2: Shared spatial fields. All spatially varying parameters are given Matérn(5/2) GP priors, as in the current model:

- $\xi(s) \sim \text{GP}$ — the tail shape, shared between the GPD (daily tail) and the implied GEV (block maxima)
- $\sigma(s, d(t))$ — GPD scale with seasonal modulation
- $\alpha_0(s) \sim \text{GP}$ — baseline wet-day probability
- $f_p(d(t), s)$ — seasonal component, borrowing strength across nearby stations

Layer 3: The GEV–GPD link. The Pickands–Balkema–de Haan theorem guarantees that if daily exceedances follow a $\text{GPD}(\sigma, \xi)$ with exceedance rate λ per block, the block maximum converges to a $\text{GEV}(\mu, \tilde{\sigma}, \xi)$ with:

$$\xi_{\text{GEV}} = \xi_{\text{GPD}}, \quad \tilde{\sigma} = \sigma \lambda^\xi, \quad \mu = u + \frac{\sigma}{\xi}(\lambda^\xi - 1). \quad (13)$$

This means the GEV parameters are *derived quantities* — not separately estimated — and the shape parameter ξ is informed by all daily tail observations, not just annual maxima.

Layer 4: Missing data imputation. Within each MCMC iteration:

1. Sample missing daily values from the posterior predictive of the hurdle model, conditional on the current seasonal–spatial parameters and the observed days at that station and its neighbours.
2. Recompute block maxima from observed + imputed daily values. Some imputed values may exceed the observed annual maximum, properly reflecting the uncertainty from incomplete records.
3. Evaluate the GEV likelihood (or derive return levels directly from the augmented daily series).

A.3 Implementation plan

1. **Phase 1: Single-station prototype.** Fit the hurdle + GPD model to Málaga Aeropuerto (66 years, 23 190 daily obs) in Stan. Validate the seasonal cycle and compare the implied GEV against the current station-level MLE.
2. **Phase 2: Verify the GEV–GPD link.** From the fitted daily model, derive GEV parameters via (13) and compare with existing return level estimates.
3. **Phase 3: Multi-station spatial model.** Extend to all 127 stations with shared GP priors on $\xi(s)$, $\sigma(s)$, and the seasonal parameters.
4. **Phase 4: Missing data integration.** Impute missing days within the MCMC loop; compare return levels with and without imputation.
5. **Phase 5: Non-stationarity.** Add a time trend to $\mu(t)$ or $\sigma(t)$ to model climate change effects on extremes.

Design Note: The daily model is estimated independently of (or jointly with) the GEV — the dependency is one-directional. The daily process has $\sim 1\,000\,000$ observations to learn from, so allowing the GEV to feed back into the daily model (a fully joint approach) would provide only marginal additional information. The modular approach is therefore a good approximation and substantially simpler to implement.

References

- Hazra, A., Huser, R., and Jóhannesson, Á. V. (2023). Bayesian latent Gaussian models for high-dimensional spatial extremes. In Hrafnkelsson, B., editor, *Statistical Modeling Using Bayesian Latent Gaussian Models*, chapter 7, pages 219–246. Springer.
- Hrafnkelsson, B., editor (2023). *Statistical Modeling Using Bayesian Latent Gaussian Models: With Applications in Geophysics and Environmental Sciences*. Springer.
- Hrafnkelsson, B., Siegert, S., Huser, R., Bakka, H., and Jóhannesson, Á. V. (2021). Max-and-Smooth: estimation of multivariate latent Gaussian models using the Max-and-Smooth algorithm. *Bayesian Analysis*, 16(4):1–29.
- Lindgren, F., Rue, H., and Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B*, 73(4):423–498.
- Rue, H., Martino, S., and Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B*, 71(2):319–392.
- Simpson, D., Rue, H., Riebler, A., Martins, T. G., and Sørbye, S. H. (2017). Penalising model component complexity: a principled, practical approach to constructing priors. *Statistical Science*, 32(1):1–28.
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., and Riddell, A. (2017). Stan: a probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32.
- Fuglstad, G.-A., Simpson, D., Lindgren, F., and Rue, H. (2019). Constructing priors that penalize the complexity of Gaussian random fields. *Journal of the American Statistical Association*, 114(525):445–452.
- rOpenSpain (2026). *climaemet: Climate AEMET Tools for R*. R package version 1.4.0, <https://ropenspain.github.io/climaemet/>.
- Vías, J. M., Andreo, B., Perles, M. J., Carrasco, F., Vadillo, I., and Jiménez, P. (2006). Proposed method for groundwater vulnerability mapping in carbonate (karstic) aquifers: the COP method. *Hydrogeology Journal*, 14(6):912–925.